P-C-440C
July 18, 1972
SUPERSEDING
Int. Fed. Spec. P-C-00440B(GSA-FSS)
February 20, 1963

FEDERAL SPECIFICATION

CLEANING COMPOUND, SEPTIC TANK AND DRAINS (ENZYME-BACTERIA TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers an enzyme-bacteria cleaning compound for use in septic tanks and for maintenance cleaning of drains.

1.2 Classification. The compound covered by this specification shall be one grade only.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-723 - Drums, Fiber.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 313 - Symbols for Packages and Containers for Hazardous Industrial Chemicals and Materials.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7930

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Laws and Regulation:

21 CFR-191 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

Microbiology Laboratory Guidebook

(Application for copies should be addressed to the U.S. Department of Agriculture, Consumer and Marketing Service, Technical Services Division, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society For Testing and Materials (ASTM) Standard:

ASTM E70 - Method for Determination of the pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The compound shall be a uniform free-flowing powdered or granular material consisting of active enzymes combined with bacterial culture materials. The compound may include nonionic wetting agents, emulsifying agents, active dried yeasts, urea, anhydrous dextrose and buffering salts.

3.2 Starch liquefying ability. When tested as specified in 4.2.1, the compound shall have a flow-through time of no more than 11.5 seconds.

3.3 Protein digesting ability. When tested as specified in 4.2.2, the compound shall contain no less than 15,000 proteolytic/casein units per gram (P/C units/g.)

3.4 Anaerobic bacteria. When tested as specified in 4.2.3, the compound shall develop 1.5 billion colonies per gram.

3.5 pH. When tested as specified in 4.2.4, the compound shall have a pH in the range 6.0-8.0.

3.6 Salmonella. When tested as specified in 4.2.5, the compound shall be free from salmonella bacteria.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Analytical procedure.

4.2.1 Starch liquefying test.

4.2.1.1 Reagent. Prepare a buffered starch substrate by mixing 2 percent by weight sodium bicarbonate ($NaHCO_3$) and 1 percent by weight disodium phosphate (Na_2HPO_4) with gelatinized wheat paste 1/. This should be done by admixing the dry powders, which are then stored in a wide mouth screwcap bottle.

4.2.1.2 Equipment. An electric mixer with rheostat control, a strainer with 1.5-2.5 mm. mesh, similar to a tea strainer, a 58° short-stemmed fast filtering funnel, having a top I.D. of 75 mm., an orifice diameter of 4 mm., and a stem length of approximately 75 mm., and a stopwatch.

4.2.1.3 Procedure. Weigh out 1.00 g. of compound into a 250 ml. beaker set in a water bath maintained at 58° to 60°C. Add 100 ml. of 60° ± 0.5°C. distilled water and stir slowly for 5.0 ± 0.3 minutes. At the end of this time, increase the stirrer to maximum speed (about 1700 rpm) and dump in 20 g. of reagent all at once. The reagent will immediately disperse and begin to set up. Reduce the speed of stirring as necessary to avoid splashing, and stir for 5.0 ± 0.1 minutes.

Set up the funnel and strainer in the sink, and preheat by running hot water (50° to 75°C.) through them. The strainer should be nesting in the cone of the funnel to keep lumps of starch from plugging the funnel.

At the end of the 5.0 ± 0.1 minute reaction period, pour the mixture through the funnel as quickly as possible without spilling. Before pouring, turn off the hot water used for preheating and let it drain out of the funnel.

The time measured is from the start of pouring until the last of the mixture leaves the cone of the funnel.

4.2.2 Protein digesting test.

4.2.2.1 Reagents. The following reagents will be required:

(1) Acid-alcohol solution - mix 25 ml. glacial acetic acid, 225 ml. distilled water, and 350 ml. ethanol.
(2) pH 8.0 buffer - dissolve 15.4 g. disodium phosphate (Na_2HPO_4), 15.0 g. sodium citrate, and 10.3 g. boric acid in 500 ml. distilled water. Adjust the pH to 8.0 with dilute sodium hydroxide.
(3) Substrate - prepare a 2-percent stock solution of casein by dissolving 5.0 g. casein in 200 ml. of buffer solution. Heat in water bath at 90°C. with occasional stirring until casein dissolves (2-3 minutes), cool to 40°C. and bring to 250 ml. total. This solution will keep 5 days under refrigeration, but must be reheated to redissolve before diluting to 0.2-percent by pipetting 10 ml. into a 100 ml. volumetric flask and making the solution up to the mark.
(4) Enzyme standard - a bacterial enzyme standard may be prepared by purchasing commercial material 2/ of known potency, and adjusting the potency to 15,000 P/C units per gram by mixing the material with table salt. The standard thus prepared will keep for six months if stored in the freezer.
(5) Enzyme extracts - extract 1.00 g. of enzyme standard or material to be tested in 100 ml. of water in a 250 ml. beaker. Place the beakers in a 40° ± 0.5 C. water bath for 35 to 40 minutes, with occasional stirring. At the end of that time, filter 10-20 ml. of the extract and place this filtrate in the refrigerator at 4°C. The standard enzyme should be extracted fresh each day of testing and kept in the refrigerator.

1/ Jellitac's "Clingfirm" brand has no artificial surfactant and gives consistent results.
2/ Chas. Pfizer's Maxatase is a freeze dried enzyme powder with a potency of 300,000 delft units or P/C units per gram.

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4.2.2.2 Procedure. Pipette 10 ml. of substrate into each of six 13 by 122 mm. test tubes. Chill in the refrigerator for 15 minutes and pipette 1.0 ml. of enzyme standard extract into three of them and 1.0 ml. of test material extract into the other three. Mix well and place the tubes at once into the 40° ± 0.5 C. water bath and digest. At the end of 30, 35, and 40 minutes remove a pair of tubes, one containing test material extract and one containing enzyme standard extract, and pipette 1.0 ml. of acid-alcohol solution into each tube. Agitate the tubes to ensure thorough mixing and replace in the bath for 1-2 minutes before making a visual comparison. Shake the closed tube vigorously before comparing. The tubes containing test material extract should have equal or less turbidity than those containing standard enzyme extract.

4.2.3 Anaerobic bacteria test. Weigh out 1.00 g. of compound into 50.0 ml. of sterile solution blank containing glass beads. Restopper and allow to stand for 5 minutes, then agitate vigorously for 5 additional minutes. The sample is then serially diluted, 1 to 5,000,000, using serological pipettes, and 0.50 ml. of the final dilution is added to 24.5 ml. of Brewer's Anaerobic Agar, which has been previously liquefied and cooled to 45°C. ± 0.5 C. in a cotton-stoppered 50 ml. Erlenmeyer flask. The agar is swirled carefully to mix the 0.50 ml. of sample with it while avoiding entrainment of air. The agar containing the sample is then poured into an oval culture tube, 23 x 11.5 mm. cross section, 152 mm. length, filling it to a depth of about 5 inches.

Immediately after pouring in the agar, the agar is overlaid with 18-20 mm. of sterile vaseline, paraffin or other inert sterile ingredient. The sterile sealant should be less than 55°C. at the time of pouring, and preferably as close to 50°C. as possible.

Within several minutes the liquefied agar and sample will turn yellow, indicating that a thoroughly reduced medium has been obtained. The tube is then incubated at 36° ± 1°C. for 24 hours.

Mark off the tube into 1-inch sections, count the colonies in three sections, and average them. The average count should be not less than 30 colonies to the inch, equivalent to 1.5 billion colonies per gram.

4.2.4 pH test. Dissolve 10 g. of compound in 50-60 ml. of distilled water at room temperature. Measure the pH of the solution according to ASTM Method E70.

4.2.5 Salmonella test. Salmonella bacteria shall be tested for according to Section 4.5(2) of the USDA's Microbiology Laboratory Guidebook.

4.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE I. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
	Cushioning inadequate, improper or missing.

4.3.1 Examination of cans and boxes. When packaging and packing are required to comply with PPP-B-636 or PPP-C-96, the examination of requirements shall be in accordance with the appendix of the appropriate specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A and B.

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5.1.1.1 One-ounce packet. One-ounce of the cleaning compound shall be packaged in a suitable milti-ply, heat sealable, paper and foil laminated sheet.

5.1.1.2 One or five-pound container. One pound or five pound, as specified (see 6.2) of cleaning compound shall be packaged in a can conforming to PPP-C-96, type V, class 2 exterior coating plan A.

5.1.2 Level C. Cleaning compound shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-ounce packet. Forty or eighty 1-ounce packets in accordance with 5.1.1.1 shall be packed in a class weather resistant fiberboard box conforming to PPP-B-636.

5.2.1.2 One or five-pound containers. Twenty-four filled 1-pound or six filled 5-pound containers shall be packed in accordance with level A requirements of the appendix to PPP-C-96, except for the number of cans per case.

5.2.1.3 Twenty-five or one hundred pound containers. Twenty-five or one hundred pounds of the cleaning compound shall be packed in a fiber drum conforming to type III, grade A of PPP-D-723.

5.2.2 Level B.

5.2.2.1 One-ounce packet. Forty or eighty 1-ounce packets in accordance with 5.1.1.1 shall be packed in fiberboard box conforming to PPP-B-636, class domestic.

5.2.2.2 One or five-pound containers. Twenty-four filled 1-pound or six filled 5-pound containers shall be packed in accordance with level B requirements of the appendix to PPP-C-96, except for the number of cans per case.

5.2.2.3 Twenty-five or one hundred pound containers. Twenty-five or one hundred pounds of the cleaning compound shall be packed in a fiber drum conforming to type I, grade A of PPP-D-723.

5.2.3 Level C. Cleaning compound in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Labeling and marking. The labeling and marking shall comply with Fed. Std. No. 313 and the Federal Hazardous Substance Act and Regulations Promulgated Thereunder. In addition to any special markings required in the procurement document, marking of the interior packages and exterior shipping container shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. The compound is intended for the regular maintenance of drains, traps, pipelines, and septic tanks. It is not intended for emergency cleaning of plugged drains, etc. since time is required for effective action.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of container required (see 5.1 and 5.2).
(c) Whether packing shall be in accordance with Level A, B, or C.
(d) Quantities required (see 5.2).
(e) Marking document required (see 5.3).

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